Mini-batch Gradient Descent (Motivation)

- Train a neural network
- Define a loss function
- Learn weights

Objective:

$$\theta^* = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \mathscr{L}(x_i, y_i)$$

Gradient Descent:

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{1}{N} \sum_{i=1}^{N} \nabla \mathscr{L}(x_i, y_i)$$

Mini-batch:
- Sample m examples randomly from N
- Use them to estimate gradient

Population & Sampling

Given:

- Dataset: $\{x_1, x_2, \ldots, x_N\}$

Population Mean:

$$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$$

Population Variance:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2$$

Random Sampling

- Sampling with replacement
- Each sample chosen with probability $\frac{1}{N}$

$$X_i = \begin{cases} x_1 & \text{w.p. } \frac{1}{N} \\ x_2 & \text{w.p. } \frac{1}{N} \\ \vdots & \\ x_N & \text{w.p. } \frac{1}{N} \end{cases}$$

Expectation of Sample

$$E[X_i] = \sum x_j \cdot \frac{1}{N} = \mu$$

Validation: Correct — expectation of sampled value equals population mean

Variance of Sample Mean

Sample Mean Estimator:

$$\hat{\mu} = \frac{1}{m} \sum_{i=1}^{m} X_i$$

Expectation:

$$E[\hat{\mu}] = \mu$$

Unbiased estimator

Variance:

$$\text{Var}(\hat{\mu}) = \text{Var}\left(\frac{1}{m} \sum X_i\right)$$

Using properties:

- $\text{Var}(aX) = a^2 \text{Var}(X)$
- Independence ⇒ variances add

$$= \frac{1}{m^2} \sum \text{Var}(X_i) = \frac{m\sigma^2}{m^2} = \frac{\sigma^2}{m}$$

Validation: Correct (assuming i.i.d samples)

Variance Formula (Important Identity)

$\text{Var}(X) = E[X^2] - (E[X])^2$

Standard identity — correct

SAMPLE VARIANCE ESTIMATOR

$$S^2 = \frac{1}{m-1} \sum (X_i - \hat{\mu})^2$$

Property:

$E[S^2] = \sigma^2$

Unbiased estimator

Validation: Correct (Bessel's correction used properly)

Covariance

Definition:

$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$

Equivalent:

$= E[XY] - \mu_X \mu_Y$

PROPERTIES:

1. $\text{Cov}(X, X) = \text{Var}(X)$
2. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
3. $\text{Cov}(aX, Y) = a \cdot \text{Cov}(X, Y)$
4. $\text{Cov}\left(\sum X_i, \sum Y_j\right) = \sum \sum \text{Cov}(X_i, Y_j)$

Covariance Matrix

For $X_1, X_2, \ldots, X_n$:

$M_{ij} = \text{Cov}(X_i, X_j)$

Diagonal:

diag(M) = Var$(X_1), \ldots,$ Var(X_n)

Geometric Intuition of Covariance

Positive Covariance

```
Y ↑
  |       *
  |    *
  |  *
  | *
  +----------→ X
```

Negative Covariance

```
Y ↑
  | *
  |   *
  |     *
  |        *
  +----------→ X
```

Zero Covariance

```
Y ↑
  |  *   *   *
  |    *   *
  | *    *    *
  |
  +----------→ X
```

NOTE - Covariance measures only linear dependence

Example:

- $X \in \{-1,1\}$, equal probability
- $Y = X^2$

Then:

- $Y = 1$ always
- Cov$(X, Y) = 0$

But clearly dependent!

Cov$(X, Y) = 0 \not\Rightarrow$ independence

Validation: Excellent example — conceptually important

Correlation Coefficient

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Properties:

$\rho \in [-1,1]$
$\rho = 1$ → perfect positive linear relation
$\rho = -1$ → perfect negative linear relation

Joint Gaussian Example

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right)$$

Correlation = ρ

Continuous Example

Given:

- $X \sim$ **Uniform**$(0,1)$
- $Y \mid X = x \sim$ **Uniform**$(0,x)$

Joint Density:

$$f_{XY}(x, y) = 1 \cdot \frac{1}{x}, \quad 0 < y < x < 1$$

Marginal of Y:

$$f_Y(y) = \int_y^1 \frac{1}{x} dx = -\ln y$$

Means:

$$\mu_X = \frac{1}{2}, \quad \mu_Y = \int_0^1 y(-\ln y)dy = \frac{1}{4}$$

Expectation:

$$E[XY] = \frac{1}{6}$$

Covariance:

$$\textbf{Cov}(X, Y) = \frac{1}{6} - \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{24}$$

Variances:

$$\textbf{Var}(X) = \frac{1}{12}, \quad \textbf{Var}(Y) = \frac{7}{144}$$

Correlation:

$$\rho = \sqrt{\frac{3}{7}}$$

Validation: All computations are correct

Example : Application: Feature Selection

- Features: $X_1, X_2, \ldots, X_{100}$
- Target: Y

Use:

- Correlation matrix ρ_{X_i, X_j}
- Identify:
  - Highly correlated features → redundancy
  - Features strongly correlated with Y → useful

Example:

- X_1: Study hours
- X_2 : Problem solved
- X_3 : Sleep hours
- Y : Exam marks

If: $\rho(X_1, X_2) > 0.9$, Drop one feature